

Nileet Savale

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PROFESSIONAL SUMMARY

Computer Science M.S. candidate at Indiana University, Co-Founder & CTO of **Stamp'd** (getstampd.org), and Research Assistant applying ML pipelines to large-scale **NHANES/NIH** datasets. Experienced in containerized deployments, distributed systems, CI/CD, and GPU-accelerated ML frameworks with a focus on building scalable, high-performance AI and cloud-native systems.

EDUCATION

- **Indiana University** | *Master of Science in Computer Science, GPA: 3.7* Aug 2024 – Jul 2026
- **Savitribai Phule Pune University** | *Bachelor of Engineering in Computer Science, CGPA: 8.9* Jul 2020 – Jul 2024

EXPERIENCE

- **Co-Founder & CTO** | [Stamp'd](#) - Travel-readiness platform, (AWS, FastAPI, Docker, PostgreSQL) Jan 2026 – Present
 - Architected AWS infrastructure (EC2, S3, RDS, Cognito) with **Terraform IaC**; built end-to-end **CI/CD pipeline** with container orchestration and zero-downtime deployments across web and mobile (React Native).
 - Containerized **FastAPI** microservices with Docker Compose; integrated Sherpa API across **180+ countries** with **Cognito OAuth/OIDC**, JWT auth, and multi-tenant **PostgreSQL** schema via Alembic migrations.
 - Implemented ML-based travel risk scoring and data normalization pipelines; applied distributed patterns (async queues, event-driven sync, caching) to maintain **sub-200ms API response times** at scale.
- **Research Assistant - Data Analytics & ML** | *Prof. Sixtus Augree, School of Public Health, IUB* Jan 2026 – Present
 - Replicating Kruger et al. (2004) on **NHANES 2001–2018** (9 cycles, 180K+ records); built automated CDC/NIH ingestion pipeline (12 components × 9 cycles) with CV reliability checks and structured outputs for downstream statistical modeling; configured **NIH SRA Toolkit + dbGaP** controlled-access workflows for genomic sequencing data.
 - Designed **ML modeling framework** — regression, classification, and correlation on dietary recall, metabolic biomarkers (insulin, A1C, cholesterol), and WHD080 variables at population scale to inform global health interventions.
- **CS Team Lead - Global Health Impact** | *Tobias Center, IUB* Aug 2025 – Present
 - Led **30-member** cross-functional CS team; coordinated with data team on entity mapping pipelines and with engineering sub-team to build a **global health forecasting tool** and internal intranet for team collaboration.
 - Migrated research platform from **Flask to Next.js**; architected **Jenkins CI/CD** pipelines (75% faster deploys); managed **Jetstream2 cloud VM** (99.9% uptime) and containerized Flask/Nginx infrastructure.
- **Teaching Assistant - Elements of AI** | *Prof. Saul Blanco, IUB* Aug 2025 – Dec 2025
 - Supported 80+ students with office hours and sessions on **deep learning, inference algorithms, and neural network optimization**; integrated real-world AI systems applications throughout coursework.
 - Designed supplementary problem sets focused on search algorithms, probabilistic inference, and neural architectures, reinforcing core concepts in AI systems and performance analysis.
- **Project Management Intern** | *Wipro PARI Ltd, Pune* Nov 2023 – Apr 2024
 - Contributed to FAT3 testing for 3 major industrial automation projects, achieving **15% reduction** in testing time and **20% improvement** in validation accuracy through structured test planning and defect tracking.
 - Collaborated with cross-functional engineering teams to document test workflows and coordinate validation cycles across large-scale production software deployments.

PROJECTS

- **Suits - AI Workplace Conflict Resolution** | *Next.js 14, FastAPI, PostgreSQL, Claude API, Docker* [Github](#)
 - Built multi-party AI mediation platform using **Claude Haiku** to simultaneously analyze independent submissions, classify disagreements (factual vs. values), score resolution potential, and generate structured HR mediation guides with questions.
 - Implemented deterministic **policy-based triage layer** routing cases into 3 severity tiers via keyword/regex matching; Tier 1 bypasses AI for direct HR escalation; deployed on **Jetstream** via Dokploy with witness threshold auto-triggering analysis.
- **NoteNest - AI Smart Notes** | *React.js, FastAPI, MongoDB, OpenAI API, Docker* [Github](#)
 - Implemented **streaming LLM inference** via GPT-4o-mini with **token budgeting** and latency-optimized API calls, gaining hands-on familiarity with inference throughput tradeoffs in production workloads.
 - Dockerized full-stack deployment with single-command launch; instrumented real-time analytics dashboard to monitor AI categorization accuracy and **request latency** across concurrent sessions.
- **FlashCard AI App** | *React Native, Python, FastAPI, Firebase, OpenAI API, Docker* [Github](#)
 - Integrated **LLM inference pipeline** to convert lecture PDFs into structured flashcards via OpenAI API; fully **Dockerized** backend with FastAPI, managing prompt engineering, token optimization, and async processing for batch document workloads.
 - Implemented SM-2 spaced repetition algorithm with **Firebase real-time sync** for cross-platform sessions; end-to-end deployment pipeline from PDF ingestion through **NLP inference** to mobile delivery at university scale.

TECHNOLOGIES

Languages	Python, C/C++, Java, JavaScript, TypeScript, Solidity, HTML, CSS
ML / AI	PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenAI API, NLP, Deep Learning
Inference & Sys	Docker, Kubernetes, Linux, Parallel & Distributed Systems, CUDA concepts, vLLM (familiarity)
Cloud & DevOps	AWS (EC2, S3, RDS, Cognito, CloudFront), Terraform, Jenkins, CI/CD, IaC
Frameworks & DBs	React.js, React Native, FastAPI, Node.js, Express.js, PostgreSQL, MongoDB, Firebase

PUBLICATIONS

- [Transforming Household Waste Management: IoT-Enabled Smart Dustbin](#)

IEEE, Oct 2024